

JEE MAIN REVISION MASTER: ELECTROMAGNETIC INDUCTION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 153 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Magnetic Flux & Faraday-Lenz Law	101 PYQs	Core Principle
Very High	Self Inductance & RL Transient Circuits	44 PYQs	Inductance & Circuits
Very High	Motional EMF (Translational & Rotational)	27 PYQs	Motional Induction
Very High	Mutual Inductance & Coupled Coils	27 PYQs	Coupling
Moderate	Induced Electric Field & Eddy Currents	7 PYQs	Field Theory

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Magnetic Flux & Faraday-Lenz Law	$\Phi_B = \vec{B} \cdot \vec{A} = BA \cos\theta$, $\mathcal{E} = -\frac{d\Phi_B}{dt}$, $\mathcal{E} = -N \frac{d\Phi_B}{dt}$, $q = \frac{\Delta \Phi}{R}$
Self Inductance & RL Transient Circuits	$\mathcal{E} = -L \frac{di}{dt}$, $L = \mu_0 n^2 A l$, $U = \frac{1}{2} L I^2$, $\tau = \frac{L}{R}$, $I(t) = \frac{V}{R} (1 - e^{-t/\tau})$
Motional EMF (Translational & Rotational)	$\mathcal{E} = B v L$, $I = \frac{B v L}{R}$, $F_{\text{mag}} = \frac{B^2 L^2 v}{R}$, $\mathcal{E}_{\text{rot}} = \frac{1}{2} B \omega L^2$
Mutual Inductance & Coupled Coils	$\mathcal{E}_2 = -M \frac{di_1}{dt}$, $M = \mu_0 n_1 n_2 (\pi r_{\text{inner}}^2) l$, $M = k \sqrt{L_1 L_2}$ ($0 \leq k \leq 1$)
Induced Electric Field & Eddy Currents	$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$, $E(r) = \frac{r}{2} \frac{dB}{dt}$ ($r \leq R$), $E(r) = \frac{R^2}{2r} \frac{dB}{dt}$ ($r > R$)

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Magnetic Flux & Faraday-Lenz Law (Very High Yield — 101 PYQs)

Core Principle: Magnetic flux $\Phi_B = \vec{B} \cdot \vec{A} = BA \cos\theta$. Induced EMF $\mathcal{E} = -(\mathrm{d}\Phi_B / \mathrm{d}t) = -N (\mathrm{d}\Phi_B / \mathrm{d}t)$. Lenz's law dictates that induced currents oppose the change in magnetic flux.

■ **Caution / JEE Pitfall:** 1. Total induced charge flown $\Delta q = (\Delta\Phi_B / R)$ is independent of the rate or time interval of flux change! 2. Angle θ is measured from the surface normal $\vec{A}$, not the coil plane.

■ **Speed Trick:** When an N-turn coil rotates with angular velocity ω in field B: $\mathcal{E}(t) = N B A \omega \sin(\omega t)$, with peak EMF $\mathcal{E}_0 = N B A \omega$.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "A 30 cm long solenoid has 10 turns per cm and area of 5cm^2 . The current through the solenoid coil varies from 2 A to 4 A in 3.14 s. $T = \dots$ " **[Ans: B]**
- **[2026 Evening Shift] Q#1 (MCQ):** "A circular current loop of radius R is placed inside square loop of side length L ($L \ll R$) such that they are co-planar and their cente..." **[Ans: D]**
- **[2026 Evening Shift] Q#1 (MCQ):** "Suppose a long solenoid of 100 cm length, radius 2 cm having 500 turns per unit length, carries a current $I = 10\sin(\omega t)$ A, where $\omega = \dots$ " **[Ans: N/A]**

3.2 Self Inductance & RL Transient Circuits (Very High Yield — 44 PYQs)

Core Principle: Self-inductance $L = (N\Phi / I)$, induced back-EMF $\mathcal{E} = -L(\mathrm{d}I / \mathrm{d}t)$. Solenoid $L = \mu_0 n^2 A l$. RL circuit current growth: $I(t) = (V / R)(1 - e^{-t/\tau})$ with inductive time constant $\tau = (L / R)$.

■ **Caution / JEE Pitfall:** At $t = 0^+$ (immediately after switch closing), an unenergized inductor acts as an open circuit ($R \rightarrow \infty$). At $t \rightarrow \infty$ (steady state), an ideal inductor acts as a zero-resistance short wire.

■ **Speed Trick:** Magnetic energy stored: $U_B = (1/2) L I^2$; energy density in space: $u_B = (B^2 / 2\mu_0)$. Inductors resist sudden changes in current, not voltage.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "A 30 cm long solenoid has 10 turns per cm and area of 5cm^2 . The current through the solenoid coil varies from 2 A to 4 A in 3.14 s. $T = \dots$ " **[Ans: B]**
- **[2026 Evening Shift] Q#1 (MCQ):** "Suppose a long solenoid of 100 cm length, radius 2 cm having 500 turns per unit length, carries a current $I = 10\sin(\omega t)$ A, where $\omega = \dots$ " **[Ans: N/A]**
- **[2026 Morning Shift] Q#1 (Numerical):** "Inductance of a coil with 10^4 turns is 10 mH and it is connected to a dc source of 10 V with internal resistance of 10Ω . The en..." **[Ans: 20]**

3.3 Motional EMF (Translational & Rotational) (Very High Yield — 27 PYQs)

Core Principle: Motional EMF in a conductor moving through field $\vec{B}$: $\mathcal{E} = B v L$. For a rod rotating about one pivot with angular speed ω : $\mathcal{E}_{\text{rot}} = (1/2) B \omega L^2$.

■ **Caution / JEE Pitfall:** 1. Mechanical power delivered by external pulling force $P_{\text{(mech)}} = F_{\text{(ext)}} v$ exactly matches electrical Joule dissipation $I^2 R = (B^2 L^2 v^2 / R)$. 2. For an arbitrary curved wire, replace it with the straight vector connecting its endpoints.

■ **Speed Trick:** Direction of induced EMF follows $\vec{v} \times \vec{B}$. Positive charges accumulate at the end pointed to by $\vec{v} \times \vec{B}$.

Representative Source PYQs:

- [2026 Morning Shift] Q#2 (MCQ): "A 20 m long uniform copper wire held horizontally is allowed to fall under the gravity ($g=10\text{m/s}^2$) through a uniform horizontal magnetic..." [Ans: N/A]
- [2026 Morning Shift] Q#3 (MCQ): "A 1 m long metal rod AB completes the circuit as shown in figure. The area of circuit is perpendicular to the magnetic field of 0.10 T. If ..." [Ans: B]
- [2026 Morning Shift] Q#3 (Numerical): "A simple pendulum made of mass 10 g and a metallic wire of length 10 cm is suspended vertically in a uniform magnetic field of 2 T. The mag..." [Ans: 100]

3.4 Mutual Inductance & Coupled Coils (Very High Yield — 27 PYQs)

Core Principle: Coupled flux in secondary coil $\Phi_2 = M I_1$, induced EMF $\mathcal{E}_2 = -M(dI_1 / dt)$. Coaxial solenoids $M = \mu_0 n_1 n_2 (\pi r_{\text{(inner)}}^2) l$. Coupling coefficient $M = k\sqrt{L_1 L_2}$ ($0 \leq k \leq 1$).

■ **Caution / JEE Pitfall:** For concentric coaxial coils, always use the cross-sectional area of the INNER coil ($r_{\text{(inner)}}$), because the magnetic field of the outer coil only penetrates the inner core.

■ **Speed Trick:** Reciprocity theorem holds: $M_{12} = M_{21}$ regardless of differing geometry or number of turns.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A circular current loop of radius R is placed inside square loop of side length L ($L \gg R$) such that they are co-planar and their cente..." [Ans: D]
- [2026 Morning Shift] Q#2 (Numerical): "In the given circuit below inductance values of L_1, L_2 and L_3 are same. The magnetic energy stored in the entire circuit is $\$(U_{\dots})$ " [Ans: 6]
- [2026 Morning Shift] Q#3 (MCQ): "An inductor of inductance 10 mH having resistance of 100Ω is connected to battery of E.M.F. 1.0 V through a switch as shown in the fi..." [Ans: A]

3.5 Induced Electric Field & Eddy Currents (Moderate Yield — 7 PYQs)

Core Principle: A time-varying magnetic field generates a non-conservative, circular electric field: $\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$. Inside cylinder ($r \leq R$): $E(r) = \frac{r}{2} \frac{dB}{dt}$; outside ($r > R$): $E(r) = \frac{R^2}{2r} \frac{dB}{dt}$.

■ **Caution / JEE Pitfall:** Induced electric fields are non-electrostatic: field lines form continuous closed loops and work done along a closed circuit is non-zero ($\oint \vec{E} \cdot d\vec{l} = \oint \vec{E} \cdot d\vec{l} \neq 0$)!

■ **Speed Trick:** Eddy currents and hysteresis losses in transformer cores are minimized by using thin laminated sheets of silicon steel insulated by varnish.

Representative Source PYQs:

- **[2024 Evening Shift] Q#4 (MCQ):** "A transformer has an efficiency of 80% and works at 10V and 4kW. If the secondary voltage is 240V, then the current in the secondary..." **[Ans: N/A]**
- **[2023 Morning Shift] Q#1 (MCQ):** "Certain galvanometers have a fixed core made of non magnetic metallic material. The function of this metallic material is..." **[Ans: N/A]**
- **[2023 Evening Shift] Q#2 (MCQ):** "Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R. Assertion A: A bar magnet dropped t..." **[Ans: B]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

A 30 cm long solenoid has 10 turns per cm and area of 5cm^2 . The current through the solenoid coil varies from 2 A to 4 A in 3.14 s. The e.m.f. induced in the coil is $\alpha \times 10^{-5}\text{V}$. The value α is ____.

- (A) 60
- (B) 12 **[Correct]**
- (C) 120
- (D) 34

Correct Answer: B

Solution Analysis: According to Faraday's Law, a change in the magnetic flux (Φ) through a coil induces an electromotive force (e.m.f.) in that coil. The magnitude of the induced e.m.f. (ϵ) is given by: $\epsilon = N \frac{d\Phi}{dt}$ Where N is the total number of turns. For a solenoid, the magnetic flux Φ through each turn is $\Phi = B \cdot A$, where B is the magnetic field inside the solenoid and A is the cross-sectional area. The magnetic field B inside an ideal solenoid is: $B = \mu_0 n I$ Where μ_0 is the permeability of free space ($4\pi \times 10^{-7}\text{T} \cdot (\text{m} / \text{A})$), n

Benchmark Problem 2: JEE Main 2026 (Online) 22nd January Morning Shift (Q#1)

Inductance of a coil with 10^4 turns is 10 mH and it is connected to a dc source of 10 V with internal resistance of 10Ω . The energy density in the inductor when the current reaches $\left(\frac{1}{e}\right)$ of its maximum value is $\alpha \pi \times \frac{10^2 \text{J}}{\text{m}^3}$. The value of α is ____ . ($\mu_0 = 4\pi \times 10^{-7} \text{Tm/A}$).

Correct Answer: 20

Solution Analysis: When an inductor is connected to a DC voltage source (V) with a resistance (R), it initially opposes the flow of current. Over time, the current increases until it reaches a steady, maximum state. Once the current is steady, the inductor acts like an ideal wire (zero resistance). Therefore, the maximum current (i_{max}) is determined by Ohm's Law : $i_{\text{max}} = (V / R)$ For given circuit : Voltage $V = 10\text{V}$ Resistance $R = 10\Omega$ $i_{\text{max}} = (10 / 10) = 1\text{A}$ The current in the circuit is $\left(\frac{1}{e}\right)$ time the maximum value, $i = i_{\text{max}} \times \left(\frac{1}{e}\right)$ $i = 1 \times \left(\frac{1}{e}\right)$

Benchmark Problem 3: JEE Main 2026 (Online) 5th April Morning Shift (Q#2)

In the given circuit below inductance values of L_1 , L_2 and L_3 are same. The magnetic energy stored in the entire circuit is (U_t) and that stored in the L_2 inductor is (U_l) . U_t/U_l is _____. (Ignore the mutual inductance if any)

Correct Answer: 6

Solution Analysis: The magnetic energy (U) stored in an inductor of inductance L carrying a current I is given by the formula: $U = \frac{1}{2}LI^2$. In the given circuit L_1 is in series with a parallel combination of L_2 and L_3 . Let the total current entering the circuit be I . Then the current through L_1 is $I_1 = I$. Since L_2 and L_3 are in parallel and have the same inductance ($L_2 = L_3 = L$), the current I will distribute equally between them. So, the current through L_2 is $I_2 = (I/2)$ and current through L_3 is $I_3 = (I/2)$. Using the energy formula

Benchmark Problem 4: JEE Main 2025 (Online) 23rd January Morning Shift (Q#4)

In the given circuit the sliding contact is pulled outwards such that electric current in the circuit changes at the rate of 8 A/s . At an instant when R is 12Ω , the value of the current in the circuit will be _____ A.

Correct Answer: 3

Solution Analysis: To analyze the given circuit, we can use the equation for induced electromotive force (EMF) in an inductor. The formula is: $\epsilon - (L \frac{dI}{dt}) - I \cdot R = 0$. For this circuit, we have: $\epsilon = 12\text{ V}$ (the applied EMF) $L = 3\text{ H}$ (the inductance of the inductor) $(\frac{dI}{dt}) = 8\text{ A/s}$ (the rate of change of current). Assuming the direction of current is such that the change in current is negative, substitute these values into the equation: $12 - 3 \times (-8) - I \times 12 = 0$. Solving for I (the current in the circuit): $12 + 24 - 12I = 0$ $36 = 12I$ $I = (36/12) = 3\text{ A}$. Thus, the